

Prediction Result

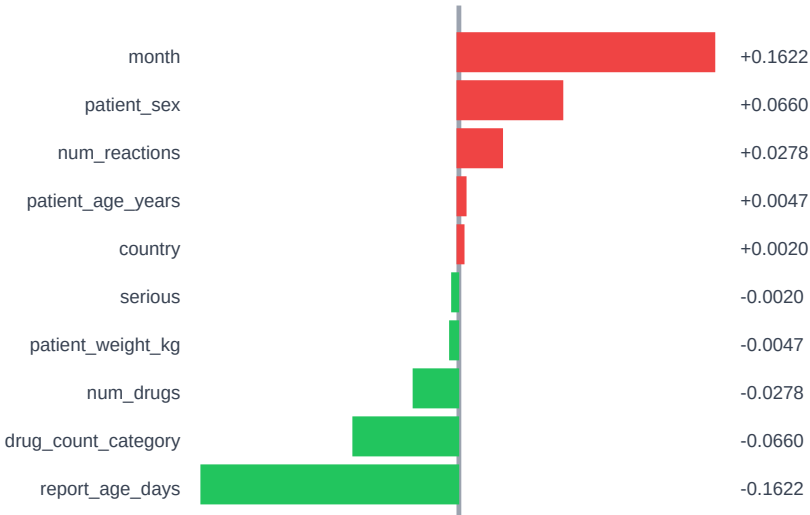
Predicted Outcome	Fatal Probability	Non-Fatal Probability	Risk Level	Severity Tier
FATAL	58.1%	41.9%	HIGH RISK	Fatal

AI Explanation (SHAP)

This risk score is driven mainly by ****500-day report lag**** (reducing fatal risk), ****month 3**** (increasing fatal risk), and ****Female sex**** (increasing fatal risk).

Feature Contributions (SHAP Values)

Red bars = increase Fatal risk Green bars = reduce Fatal risk



Feature	SHAP Value	Direction
report_age_days	-0.1622	↓ Reduces Fatal risk
month	+0.1622	↑ Increases Fatal risk
patient_sex	+0.0660	↑ Increases Fatal risk
drug_count_category	-0.0660	↓ Reduces Fatal risk
num_reactions	+0.0278	↑ Increases Fatal risk
num_drugs	-0.0278	↓ Reduces Fatal risk
patient_weight_kg	-0.0047	↓ Reduces Fatal risk
patient_age_years	+0.0047	↑ Increases Fatal risk
country	+0.0020	↑ Increases Fatal risk
serious	-0.0020	↓ Reduces Fatal risk

Input Features Used for Prediction

Feature	Value	Feature	Value
num_reactions	8	year	2023
num_drugs	12	age_group	Elderly(81+)
patient_age_years	82.0	patient_sex	Female
patient_weight_kg	68.0	drug_count_category	Polypharmacy(6+)
report_age_days	500	country	US
month	3	serious	Yes

Model Information

Algorithm	ROC-AUC	Accuracy	F1 Score (weighted)
RandomForest	0.8117	0.6758	0.7393

DISCLAIMER: This report is generated by PharmaGuard AI for research and educational purposes only. The predictions and SHAP explanations are based on historical FDA FAERS data and must NOT be used as a substitute for professional pharmacovigilance review, regulatory decision-making, or clinical medical judgment. Always consult a qualified healthcare or drug safety professional.